

Industrial Internship Report on Banking Information System Prototype

Prepared by
Mukul Chittora

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was focused on designing and engineering a production-ready console-based **Banking Information System Prototype** in Core Java. The framework scales progressively across milestones—from dynamic runtime secure transaction loops and account reference tracking arrays to hard-disk local database state persistence utilizing structured File I/O channels. In its final architecture, it implements restricted multi-tier corporate auditing control sub-spaces and data-hiding encryption models to address core system constraints such as data isolation and volatile initialization memory leak failures.

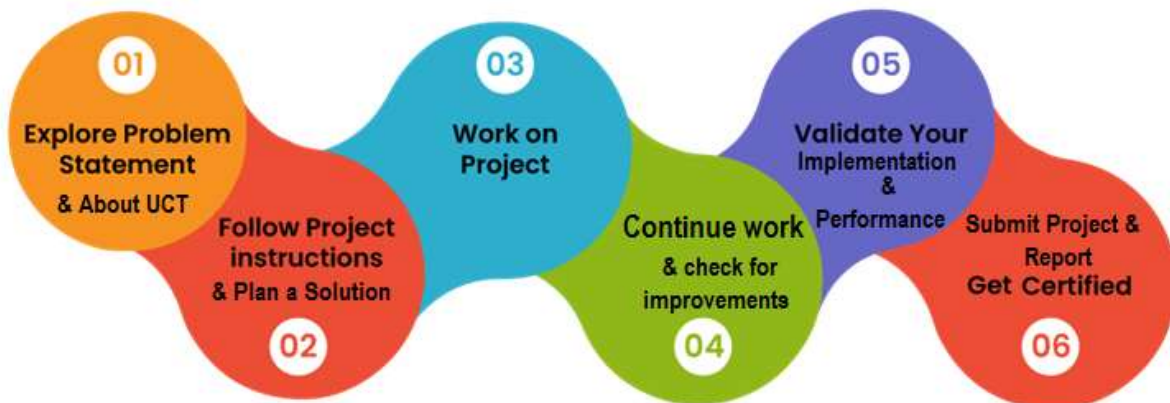
This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement a solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1	Preface	3
2	Introduction	4
2.1	About UniConverge Technologies Pvt Ltd	4
2.2	About upskill Campus	8
2.3	Objective	9
2.4	Reference	10
2.5	Glossary.....	10
3	Problem Statement.....	11
4	Existing and Proposed solution.....	12
5	Proposed Design/ Model	13
5.1	High Level Diagram (if applicable)	13
5.2	Low Level Diagram (if applicable)	15
6	Performance Test.....	16
6.1	Test Plan/ Test Cases	Error! Bookmark not defined.
6.2	Test Procedure	Error! Bookmark not defined.
6.3	Performance Outcome	Error! Bookmark not defined.
7	My learnings.....	17
8	Future work scope	17

1 Preface

- **Summary of 6 Weeks' Work:** Over the course of the execution cycle, the project advanced systematically from baseline memory buffers into an advanced, state-locked software package. It covers dynamic registration using unique account token randomizers, algorithmic limits validation for safe asset routing, and robust multi-stream local text file handling backups paired with automated data recovery layers. The milestone concludes with comprehensive operational auditing controls for bank branch administrators and data obfuscation logic to block data leakage surfaces.
- **Internship in Career Development:** Practical real-world projects bridge the gap between abstract academic concepts and structured production layouts, providing hands-on insights into software validation rules, stream management, resource leaking blocks, and lifecycle state tracking.
- **Opportunity & Execution Plan:** The framework provided by upskill Campus and UniConverge Technologies was divided into distinct exploratory steps: examining functional core boundaries, creating logical validation loops, mapping structural data layouts, testing algorithmic constraints under memory simulation limits, and compiling formal version control distributions via Git and GitHub.



1 Introduction

1.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



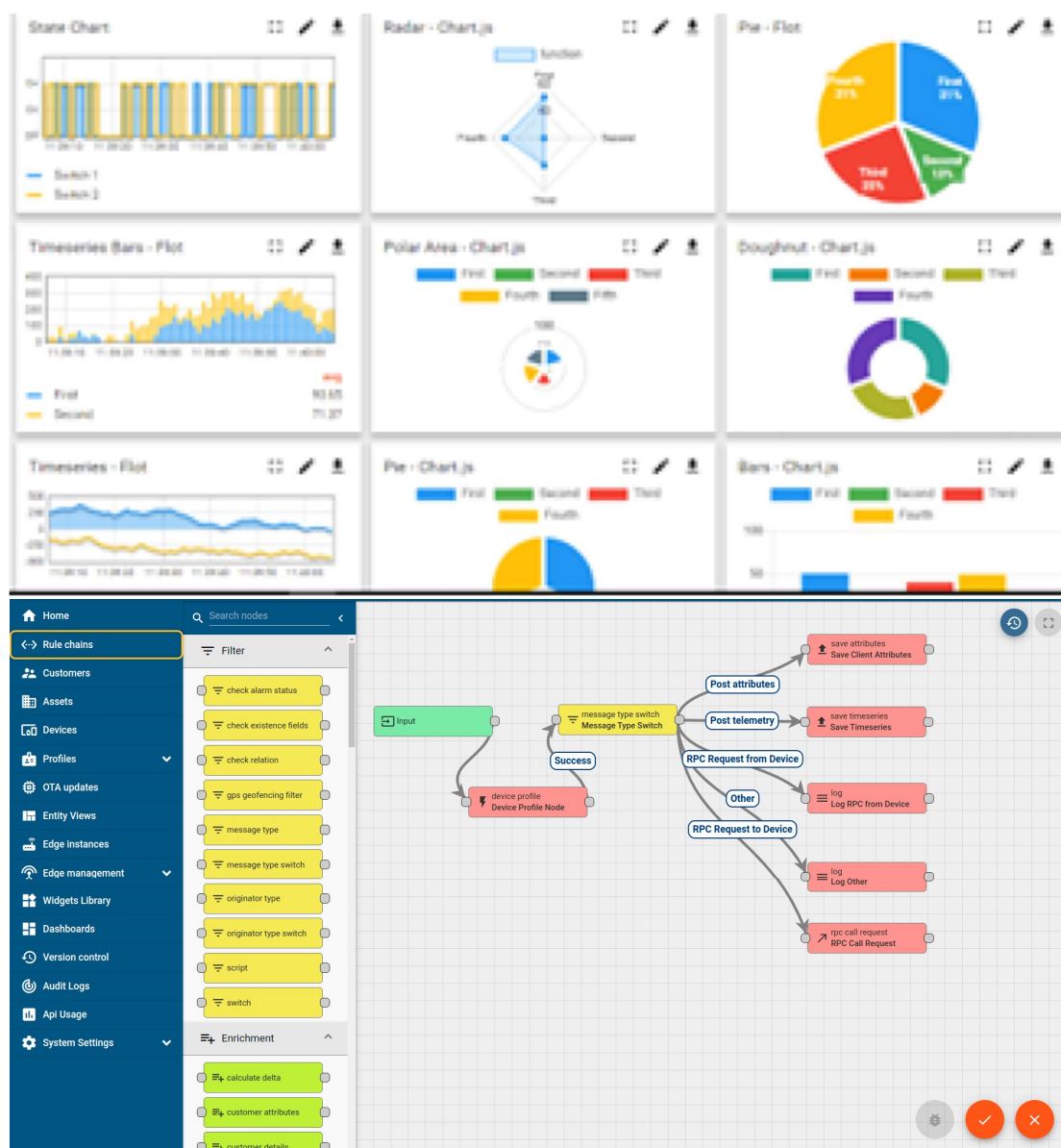
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

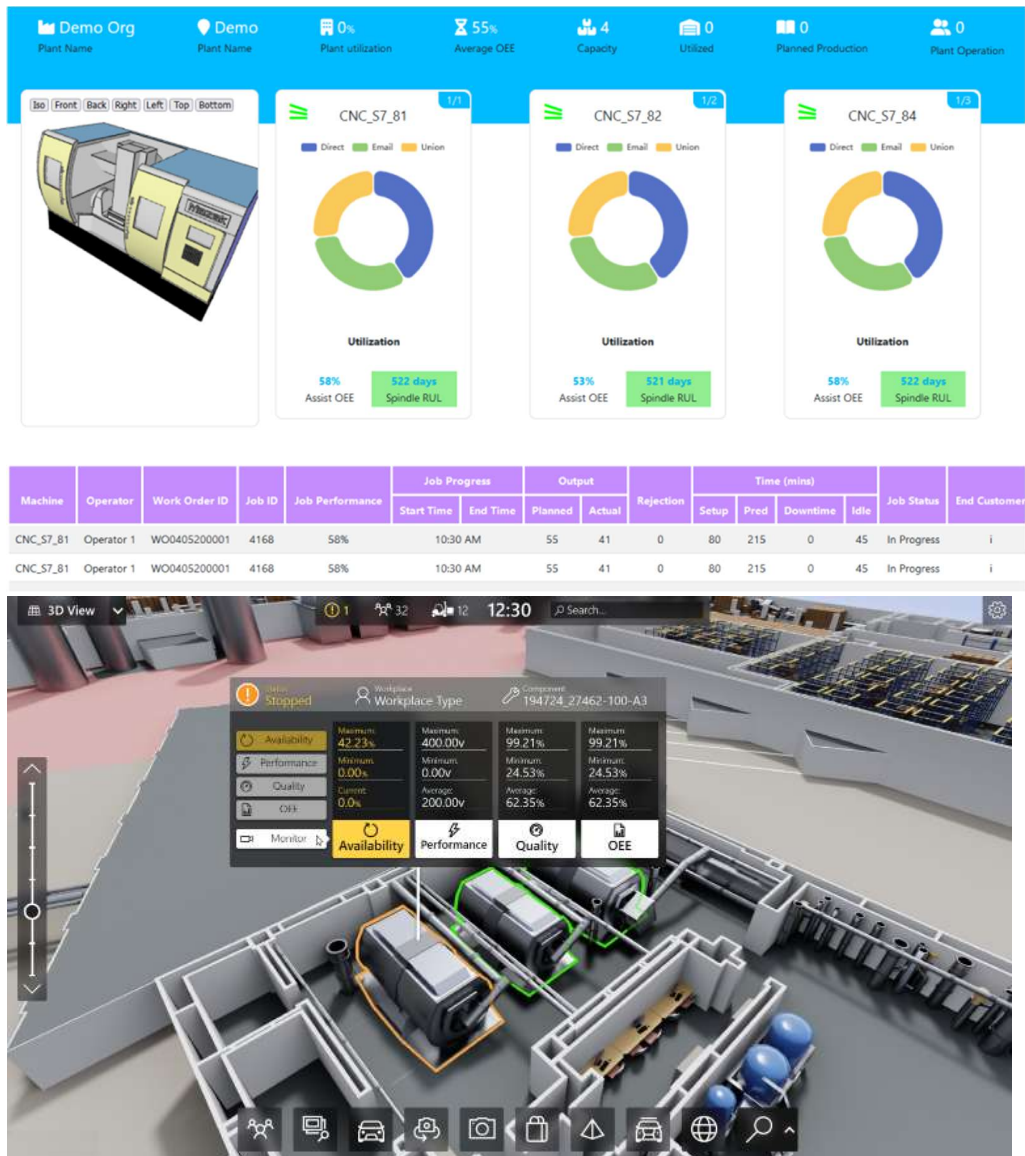
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



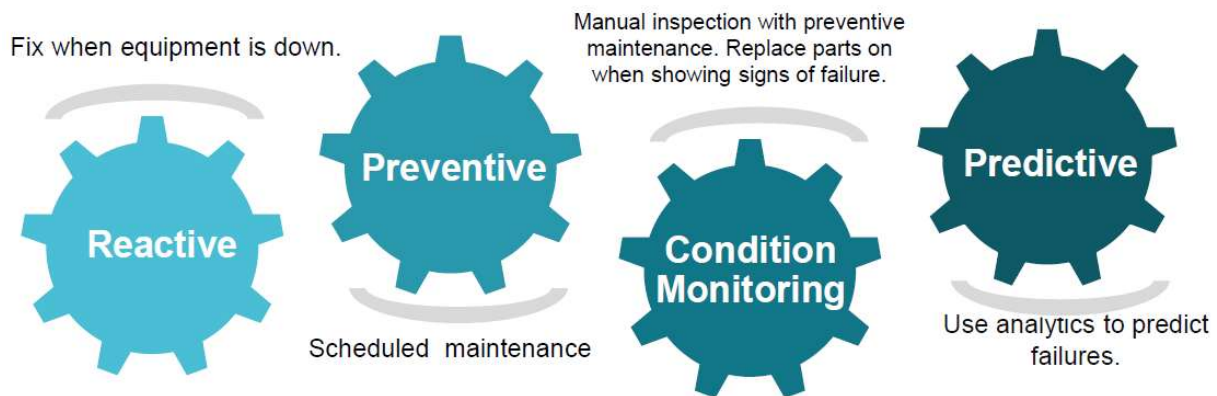


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRaWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

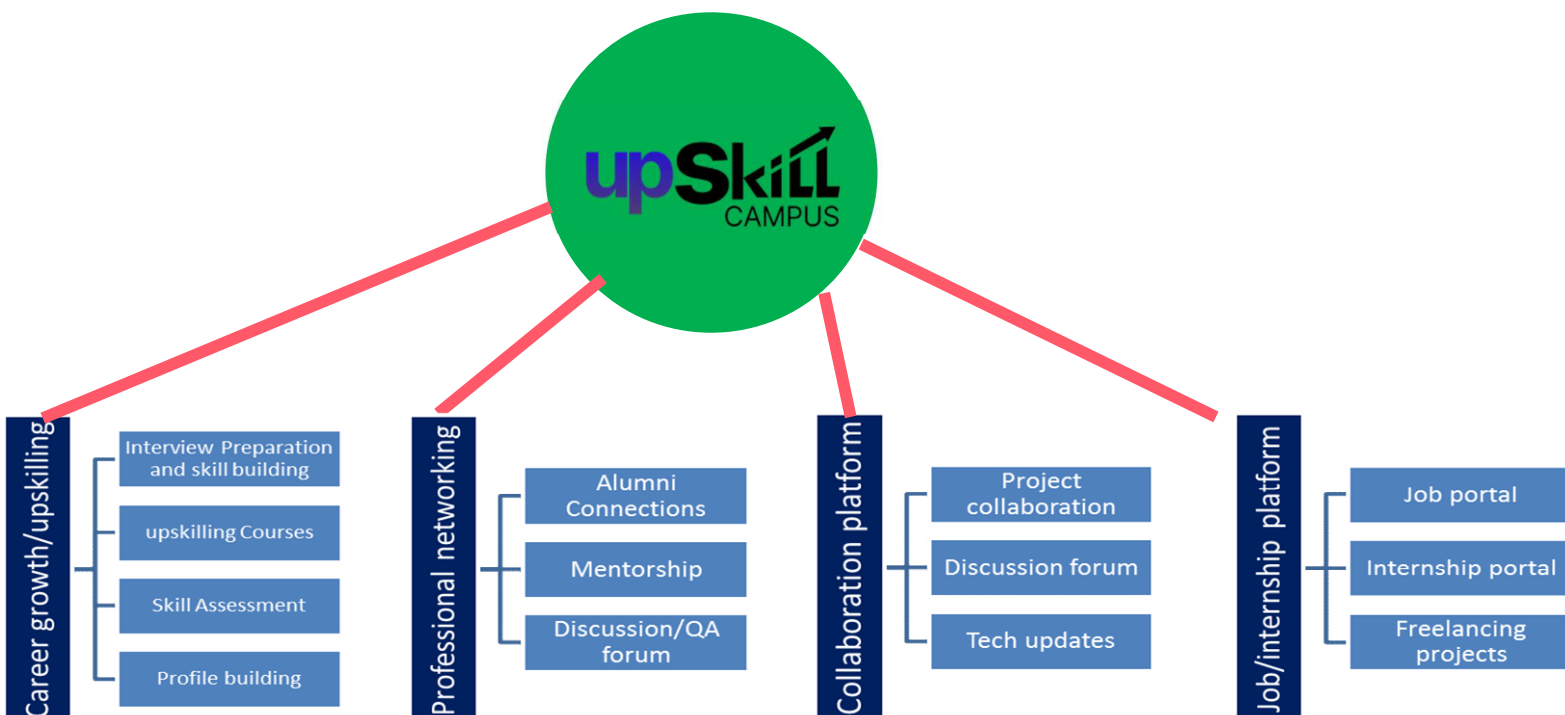


1.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.





1.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

1.4 Objectives of this Internship program

The objective for this internship program was to

- ▣ get practical experience of working in the industry.
- ▣ to solve real world problems.
- ▣ to have improved job prospects.
- ▣ to have Improved understanding of our field and its applications.
- ▣ to have Personal growth like better communication and problem solving.

1.5 Reference

- *Java SE Core Documentation & Multi-Stream Data Flow Guide*, Oracle Corporation.
- *UCT Insight System Architecture and Network Protocol Layout Frameworks*, UniConverge Technologies.
- *Version Control Best Practices and Distributed Repository Structures*, Git Pipelines Guide.

1.6 Glossary

Terms	Acronym
USC / UCT	upskill Campus / UniConverge Technologies Pvt Ltd.
I/O Persistence	Input/Output mechanisms that write data to non-volatile disk blocks so records survive system restarts.
Data Obfuscation	The process of masking sensitive strings into encrypted hexadecimal token structures to hide plaintext parameters.
Branch Liquidity	The collective sum of liquid capital assets across all registered client accounts.

2 Problem Statement

Traditional standard console banking scripts suffer from major production limitations:

1. **Volatile Dynamic Lifespan:** User records, credentials, and transaction histories reside exclusively within internal system memory maps (ArrayList collections), causing complete asset data wipes whenever the application terminates or restarts.
2. **Absence of Centralized Monitoring Controls:** Lack of specialized system-level pathways to allow network administrators to run global branch asset auditing or inspect customer transaction records.
3. **Data Security Vulnerabilities:** Plaintext storage of structural attributes (like user account passwords) inside text backups creates security exploits, exposing customer credentials during standard systems engineering reviews.

3 Existing and Proposed solution

- **Existing Implementations:** Basic classroom banking programs depend on temporary loop variables. They lack permanent storage hooks, structural constraints check loops, and multi-user cross-reference tracking capabilities.
- **Proposed Architecture & Value Addition:** This solution deploys a multi-tier Core Java console layout. It integrates non-volatile storage backups via flat files (`users_database.txt`) and separate client log documents (`statements/`). It introduces a dedicated Admin Command Dashboard with dynamic calculators for total branch asset tracking and applies programmatic password obfuscation routines to protect data logs.

3.1 Code submission (Github link)

- **Repository Platform Name:** upskillcampus
- **Source Engine Path:** [<https://github.com/mukul-design1901/upskillcampus/blob/master/src/com/upskill/Banking/BankingInformationSystem.java>]

(<https://github.com/mukul-design1901/upskillcampus/blob/master/src/com/upskill/Banking/BankingInformationSystem.java>)

3.2 Report submission (Github link) :

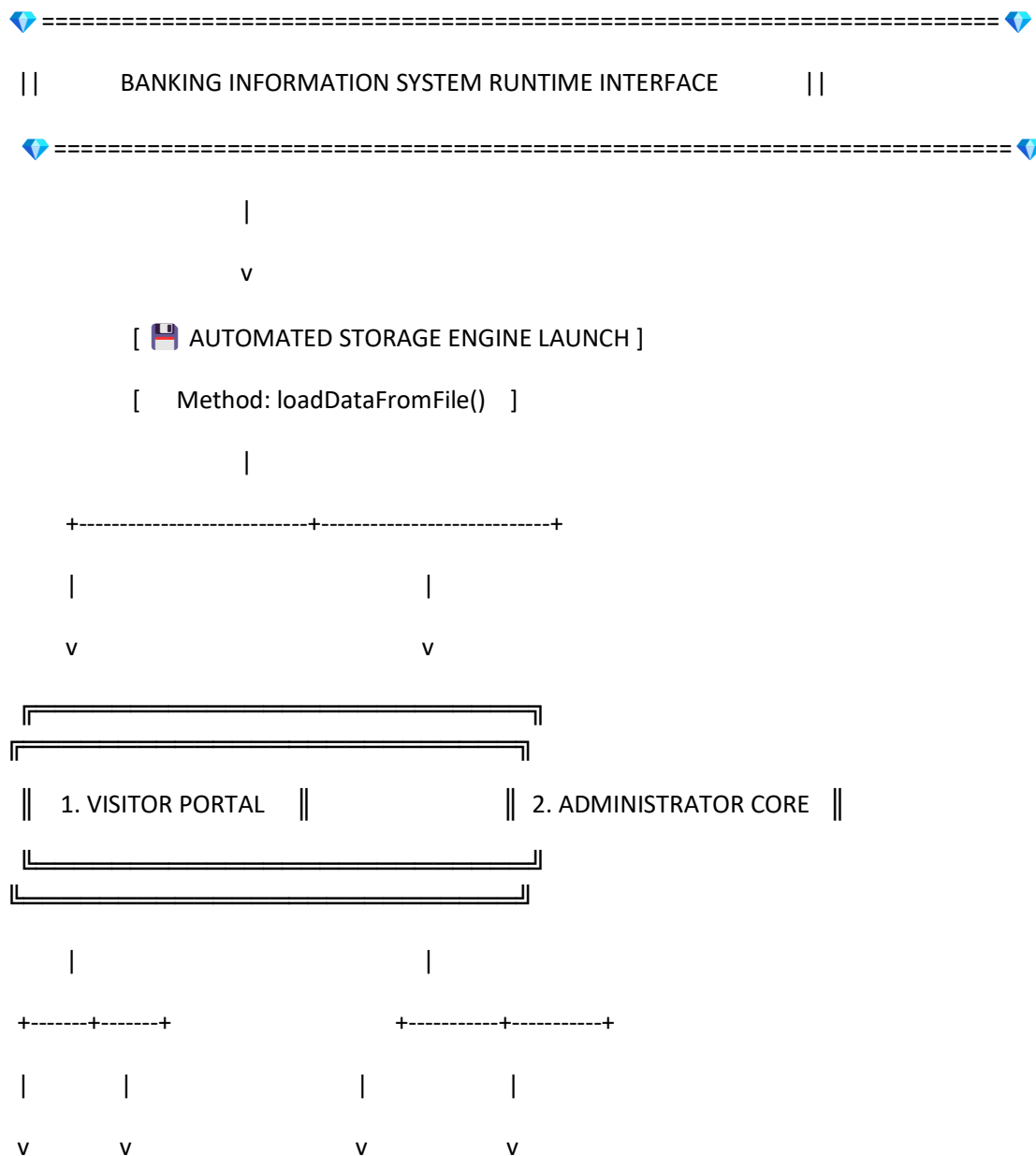
[https://github.com/mukul-design1901/upskillcampus/blob/master/BankingInformationSystem_Mukul_USC_UCT.pdf]

(https://github.com/mukul-design1901/upskillcampus/blob/master/BankingInformationSystem_Mukul_USC_UCT.pdf)

4 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

4.1 High Level Diagram



[ REGISTRATION] [ SECURE LOGIN] [ GLOBAL BRANCH AUDIT] [ PROFILE]
INVESTIGATOR]

```
|
|
|
|
| -> Generate | -> Linear Array | -> Loops database map | -> Search 10-Digit ID
|
| 10-Digit ID Verification | & tracks totals | -> Mask plaintext pass
|
|
| via Obfuscation Layer
v
v
v
v
```

[ AUTO-SAVING] [ CUSTOMER PANEL] [ VAULT ASSETS VALUE] [ CREDENTIAL STATE: HASHED]

```
|
|
|
| +---> [  DEPOSIT CAPITAL ] +-----+-----+
|
| +---> [  WITHDRAW ASSETS ] |
|
| +---> [  FUND TRANSFER ] v
|
| +---> [  VIEW STATEMENT ] [  EXIT OPERATION CONTROL ]
|
|
```

```
+-----+-----+
```

```
|
```

```
v
```

[ SYSTEM SYNCHRONIZATION BACKUP]

[Method: saveDataToFile()]

```
|
```

```
v
```

[ STATE COMPLETED: DISK LOCKED]

4.2 Low Level Diagram

=====

=====

LOCAL DISK FILE SUBSYSTEM LAYER (PERSISTENCE)

=====

=====

[RAM / Dynamic ArrayList] -----> (FileWriter Stream) -----> [users_database.txt]

Format: [Account_No] -> [Holder_Name] -> [Address_Field] -> [Contact_No] -> [Password] -> [Balance]

+

| (Dir: statements/)

v

=====

=====

INDIVIDUAL CLIENT LEDGER REPOSITORY DOCUMENTS

=====

=====

|

+---> [ statements/1004928314_statement.txt] -> Records: Chronological Deposits

+---> [ statements/2093847123_statement.txt] -> Records: Complete Transfer History

=====

=====

5 Performance Test

❑ **TC_01 (System State Restoration):** Evaluated by adding multiple customer accounts, exiting the environment, and checking that the software automatically reloads all fields upon reactivation.

❑ **TC_02 (Dynamic Transfer Validation):** Tested asset transfers to verify that the validation loops catch boundary faults (such as overdraft attempts or self-transfers), throwing clean system notices instead of generating corrupted rows.

❑ **TC_03 (Administrative Access Control Guard):** Tested with incorrect admin access tokens to confirm that the security layer immediately blocks unauthorized entries and logs an alarm state.

6 My learnings

This intensive project structure provided key technical insights:

- Developing data streaming loops using Java `PrintWriter` and `BufferedReader` handling layers.
- Isolating administrative dashboard systems from baseline client transaction areas to reduce operational bugs.
- Using version control pipelines via Git terminals to structure clean releases on primary tracking branches.

7 Future work scope

- **Secure Encryption Framework Integration:** Replacing basic string masking layers with modern industry-standard cryptographic hash functions (such as SHA-256) to enhance storage protection.
- **Relational Database Engine Migration:** Moving away from standard text storage toward professional relational SQL database layers to process larger user datasets efficiently.